

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I K Sathwik Kumar (192425051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sathwik Kumar

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Introduction

The AI-Based Student Support Assistant is designed to provide instant responses to student queries related to academic and campus services. The chatbot is developed using **Google Dialogflow**, which applies **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)** to understand user queries and generate appropriate responses. It minimizes manual effort, reduces response time, and improves the overall efficiency of student support services.

Problem Formulation

The chatbot is designed to solve the problem of delayed responses to student queries. Instead of visiting different departments, students can interact with the chatbot and receive immediate information.

The chatbot mainly supports:

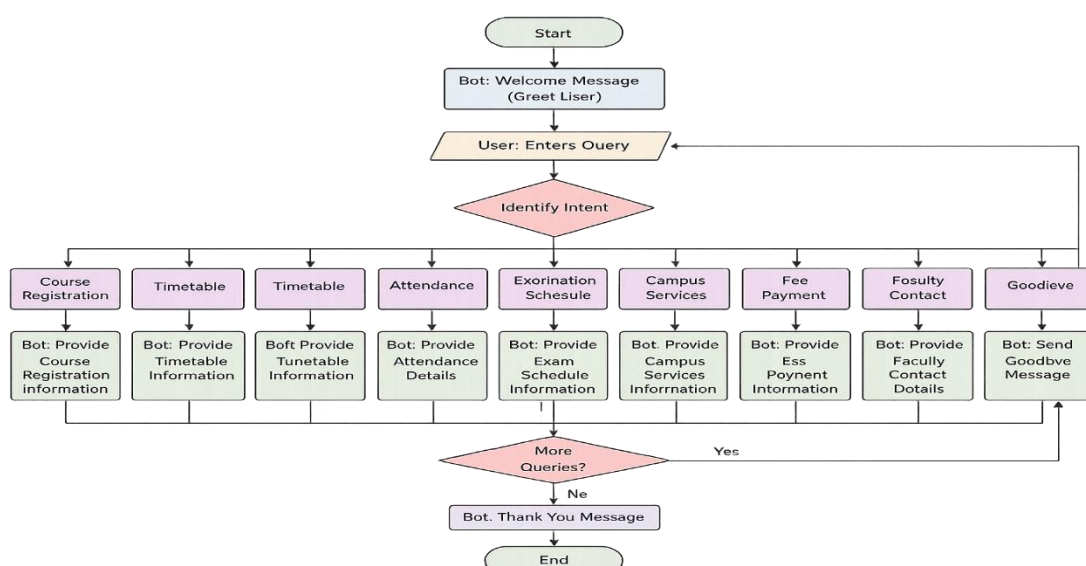
- Course Registration
- Timetable Information
- Attendance Details
- Examination Schedule
- Campus Services
- Fee Payment
- Faculty Contact Information

Initial State

The conversation starts when the student opens the chatbot. Initially, the chatbot loads all the predefined intents, entities, and responses available in Dialogflow. A welcome message is displayed, and the chatbot waits for the user's query.

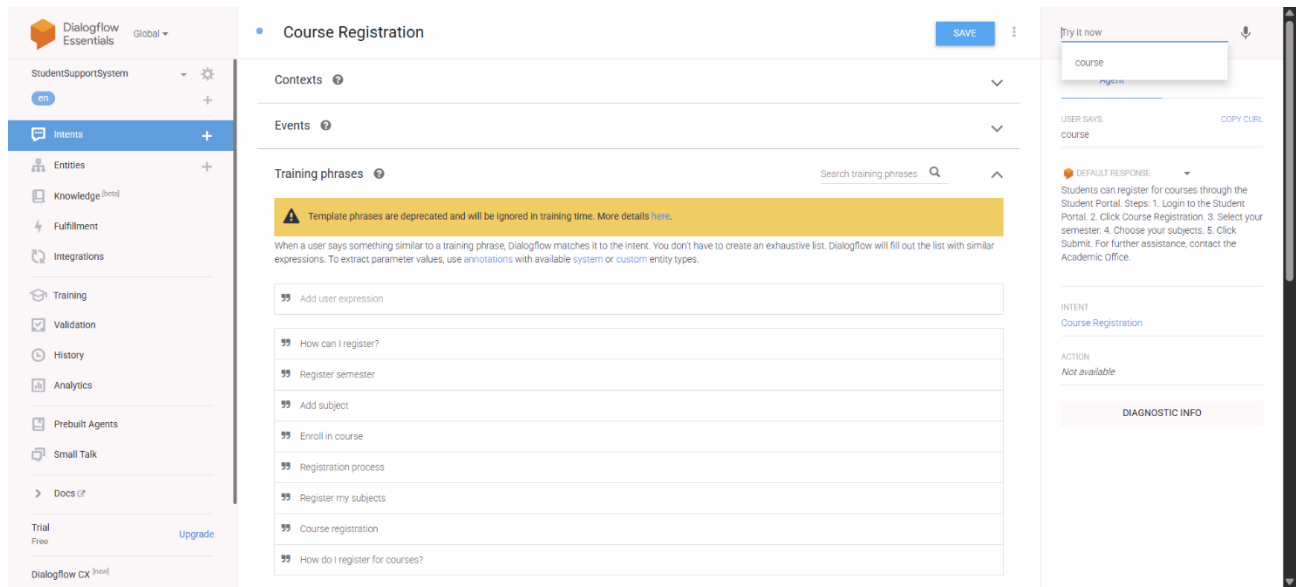
Initial State Activities:

- Chatbot starts successfully.
- Welcome message is displayed.
- User enters a query.



User Goal

The main objective of the student is to obtain accurate and quick information without manual assistance. The chatbot identifies the user's intent and provides the required information based on the available knowledge.



User Goals include:

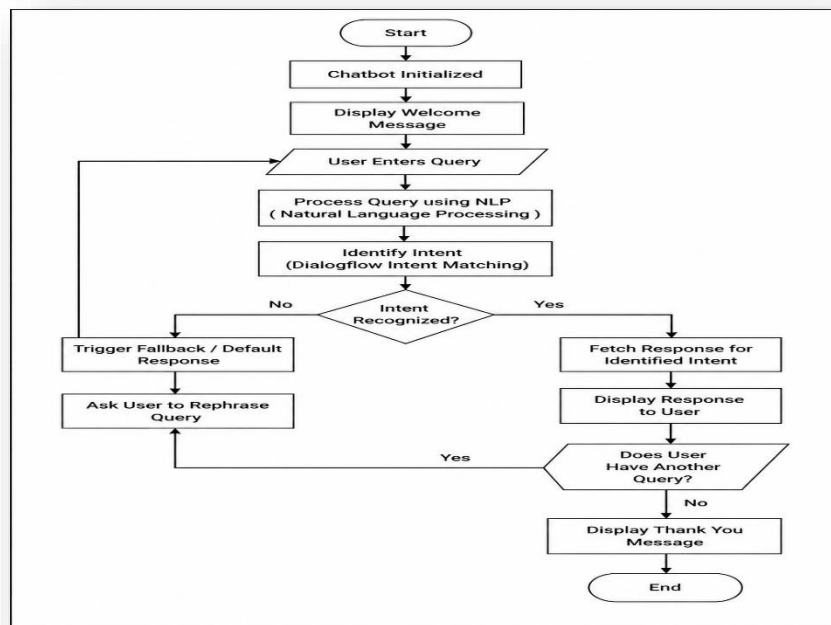
- Obtain academic information quickly.
- Access campus-related services.
- Reduce waiting time.
- Receive accurate responses.

Possible Actions

After receiving the user's query, the chatbot performs several intelligent actions before generating the response.

Actions Performed by the Chatbot:

- Accepts the user's query.
- Applies Natural Language Processing (NLP).
- Identifies the corresponding intent.
- Retrieves the appropriate response.
- Displays the response to the user.



State Transition

The chatbot continuously changes its state during the conversation. After processing each request, it either continues the interaction or terminates the conversation.

State Transition Process:

- **Start State** → Chatbot is initialized.
- **Input State** → User enters a query.
- **Processing State** → NLP identifies the intent.
- **Response State** → Appropriate response is displayed.
- **Decision State** → Checks whether another query exists.
- **End State** → Displays thank-you message and ends the session.

Goal Completion Conditions

The conversation is considered successful when the chatbot answers the student's query correctly and the user confirms that no further assistance is required.

Goal Completion Conditions:

- User receives the correct response.
- Required information is displayed successfully.
- User has no additional queries.
- Chatbot displays a thank-you message.
- Conversation ends successfully.

Intent Name	Sample User Query	Chatbot Response
Course Registration	How do I register for courses?	Provides course registration procedure.
Timetable	Show my timetable	Displays timetable information.
Attendance	What is my attendance?	Provides attendance details.
Examination Schedule	When are my exams?	Displays examination schedule.
Campus Services	What facilities are available?	Provides campus service information.
Fee Payment	How can I pay fees?	Explains fee payment procedure.
Faculty Contact	Give faculty contact details	Displays faculty contact information.

Conversation Flow Summary

The conversation flow follows a systematic approach to ensure smooth interaction between the student and the chatbot.

Flow Sequence:

1. Start the chatbot.
2. Display the welcome message.
3. Receive the user's query.
4. Identify the user's intent.
5. Generate the appropriate response.
6. Display the requested information.
7. Ask whether another query is required.

Current State	User Action / Input	System Processing	Next State
Start	User opens the chatbot	Initializes Dialogflow agent and loads intents	Welcome State
Welcome State	Chatbot displays greeting	Waits for user query	User Input State
User Input State	User enters a query	Sends query to NLP engine	Intent Recognition State
Intent Recognition State	Query is analyzed	Identifies the matching intent	Response Generation State
Response Generation State	Intent is recognized	Retrieves the appropriate response	Display Response State
Display Response State	Chatbot displays information	Asks if another query is required	Decision State
Decision State	User selects Yes	Accepts another query	User Input State
Decision State	User selects No	Displays thank-you message	End State
End State	Conversation completed	Terminates chatbot session	Goal Achieved

AI Component	Purpose in Chatbot
Natural Language Processing (NLP)	Understands the user's query.
Intent Recognition	Identifies the user's intention.
Training Phrases	Helps Dialogflow learn different query patterns.
Entities	Extracts important information from user input.
Responses	Generates appropriate replies.
Fallback Intent	Handles unknown or unsupported queries.

Intent Processing

Intent recognition is one of the most important functions of the chatbot. Each user query is compared with the predefined training phrases stored in Dialogflow. If a matching intent is found, the chatbot immediately retrieves the corresponding response. The chatbot supports various student services such as course registration, timetable information, attendance details, examination schedules, campus services, fee payment, and faculty contact information. This process enables the chatbot to understand different ways of asking the same question while maintaining high response accuracy.

Decision-Making Process

The chatbot follows a simple decision-making process throughout the conversation. After generating a response, it asks whether the user has another query. If the user wishes to continue, the chatbot returns to the input stage and processes the next request. If no further assistance is required, the chatbot displays a thank-you message and ends the conversation. This continuous decision process allows multiple queries to be handled within a single conversation, making the interaction more efficient and convenient.

Response Management

The chatbot generates responses based on the intent identified by Dialogflow. Since all responses are predefined within the Dialogflow agent, the information provided is consistent and reliable. If the chatbot cannot recognize a user's query, it activates the Default Fallback Intent and requests the user to rephrase the question. This mechanism improves the chatbot's reliability and ensures that unsupported queries are handled gracefully instead of producing incorrect responses.

Conclusion

The AI-Based Student Support Assistant developed using Google Dialogflow successfully demonstrates how Artificial Intelligence can improve student support services in educational institutions. The chatbot uses Natural Language Processing (NLP) to understand user queries, identify the appropriate intent, and generate accurate responses for services such as course registration, timetable information, attendance details, examination schedules, and campus services.

The designed conversation flow, problem formulation, and state transition model enable the chatbot to interact with users in a structured and efficient manner. By automating frequently asked queries, the system reduces response time, minimizes the workload of administrative staff, and provides students with quick and reliable information.

Overall, the proposed chatbot is a user-friendly, efficient, and scalable solution that enhances communication between students and the university. The project demonstrates the practical application of Artificial Intelligence in solving real-world educational problems and provides a strong foundation for developing more advanced intelligent student support systems in the future.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.